

questions

- what order is more **realistic**? Seems that host movement and infection should happen simultaneously but not sure how to describe that. And may need more discussion on the definition of $B(?)$

1 description

Let p denote fraction of infected patches and fraction of uninfected patches is $1 - p$.

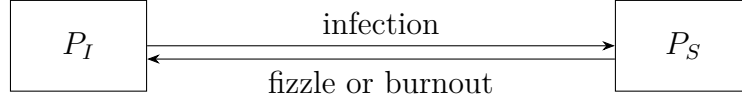


Figure 1: patch state transition

Events that can cause state change:

- **infection:** $\Delta p = Bp(1 - p)$
assuming constant B
- **fizzle or burnout:** $\Delta p = -(1 - P_1)p$
where $P_1 = P_1(R_0, \varepsilon, k = 1, N_I)$ is persistence probability.
- **migration:**

$$\begin{aligned}
 N_S &\leftarrow N_S - cp(N_S - N_I) \\
 N_I &\leftarrow N_I + c(1 - p)(N_S - N_I)
 \end{aligned}$$

individual rats move with probability c and land in patches proportional to their availability.

- **epidemic:** $N_I \leftarrow (1 - zD)N_I$
where $z = z(R_0)$ is the final size and D is the probability that an infected host dies from pathogen.
- **birth:**

$$\begin{aligned}
 N_S &\leftarrow N_S + rN_S(1 - \frac{N_S}{K}) \\
 N_I &\leftarrow N_I + rN_I(1 - \frac{N_I}{K})
 \end{aligned}$$

- **change of average population size when some patch state changes:** just calculate the average again after change of patch state. That is,

when $\Delta p > 0$,

$$N_I \leftarrow \frac{pN_I + \Delta p N_S}{p + \Delta p}$$

and N_S does not change.

when $\Delta p < 0$

$$N_S \leftarrow \frac{(1-p)N_S - \Delta p N_I}{1-p-\Delta p}$$

and N_S does not change.

To be more organized, we denote infection event with f_i , epidemic and burnout with f_e , migration with f_m , growth with f_g , then we have

$$\begin{aligned} f_m(p, N_I, N_S) &= \begin{pmatrix} p \\ N_I + c(1-p)(N_S - N_I) \\ N_S - cp(N_S - N_I) \end{pmatrix} \\ f_i(p, N_I, N_S) &= \begin{pmatrix} p + B(N_I, N_S)p(1-p) \\ \frac{N_I + B(1-p)N_S}{1 + B(1-p)} \\ N_S \end{pmatrix} \\ f_e(p, N_I, N_S) &= \begin{pmatrix} P_1(N_I)p \\ (1-zD)N_I \\ \frac{(1-p)N_S + (1-zD)(1-P_1)pN_I}{1-P_1p} \end{pmatrix} \\ f_g(p, N_I, N_S) &= \begin{pmatrix} p \\ N_I + rN_I \left(1 - \frac{N_I}{K}\right) \\ N_S + rN_S \left(1 - \frac{N_S}{K}\right) \end{pmatrix} \\ (p_{t+1}, N_{I_{t+1}}, N_{S_{t+1}}) &= f(p_t, N_{I_t}, N_{S_t}) \end{aligned}$$

$$f = f_g \circ f_e \circ f_i \circ f_m$$

2 range of parameters

Consider quadratic function $y = x + kx(1-x)$ where $0 \leq x \leq 1$ and $k \geq 0$,

$$f(x) = x + kx(1-x) = -kx^2 + (k+1)x, x \in R$$

We want to make sure that $f([0, 1]) \subset [0, 1]$. The function attains its maximum at $x^* = \frac{k+1}{2k}$. Note that $f(0) = 0$ and $f(1) = 1$, so we'll need to require that $x^* \geq 1$, i.e. $0 \leq k \leq 1$.

- **infection:** $p \leftarrow p + Bp(1-p)$, we require $0 \leq B \leq 1$
- **burnout:** $p = P_1p$, $0 \leq P_1 \leq 1$
- **epidemic:** $0 \leq z \leq 1$ and $0 \leq D \leq 1$

- birth:

$$\begin{aligned}
 N &\leftarrow N + rN(1 - \frac{N}{K}), 0 \leq N \leq K \\
 \iff x &\leftarrow x + rx(1 - x), 0 \leq x \leq 1 \\
 &\Rightarrow 0 \leq r \leq 1
 \end{aligned}$$

When there's no infection, $(0,0)$ is an unstable equilibrium and (K,K) is the only stable equilibrium, since maximum point is always greater than intersect when $r \in [0, 1]$.

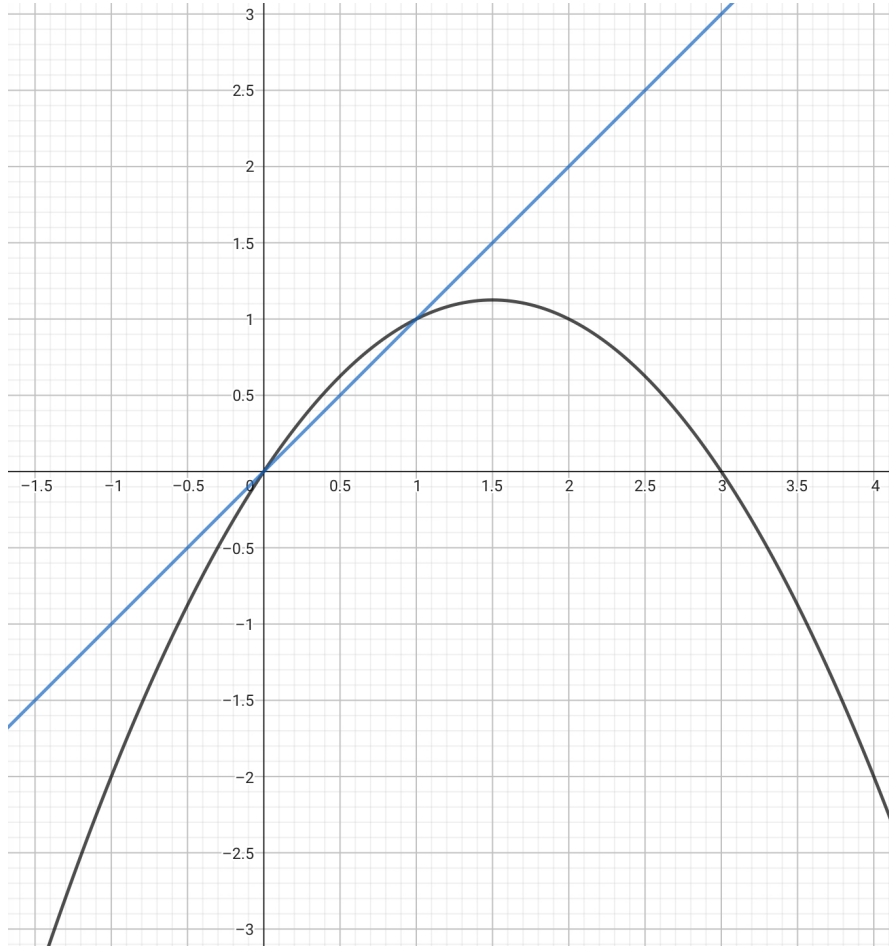


Figure 2: logistic growth

3 analysis

If we start at a point where $p > 0, N_I > 0, N_S > 0$, it's obvious that all variables remain positive during the iterations as each step maps positive values to a positive values.

We want to see what happens when we start on the axes. If we set $p = 0$ or $N_I = 0$ then population size ends up stable at carrying capacity K . $N_I = N_S = 0$ means no hosts at all so nothing would happen. It's a little weird to set $N_S = 0$ since it indicates that all hosts are infected, but then some would recover and the system would undergo burnout and

colonization, returning to the situation that $p > 0, N_I > 0, N_S > 0$, so we don't get anything special about that.

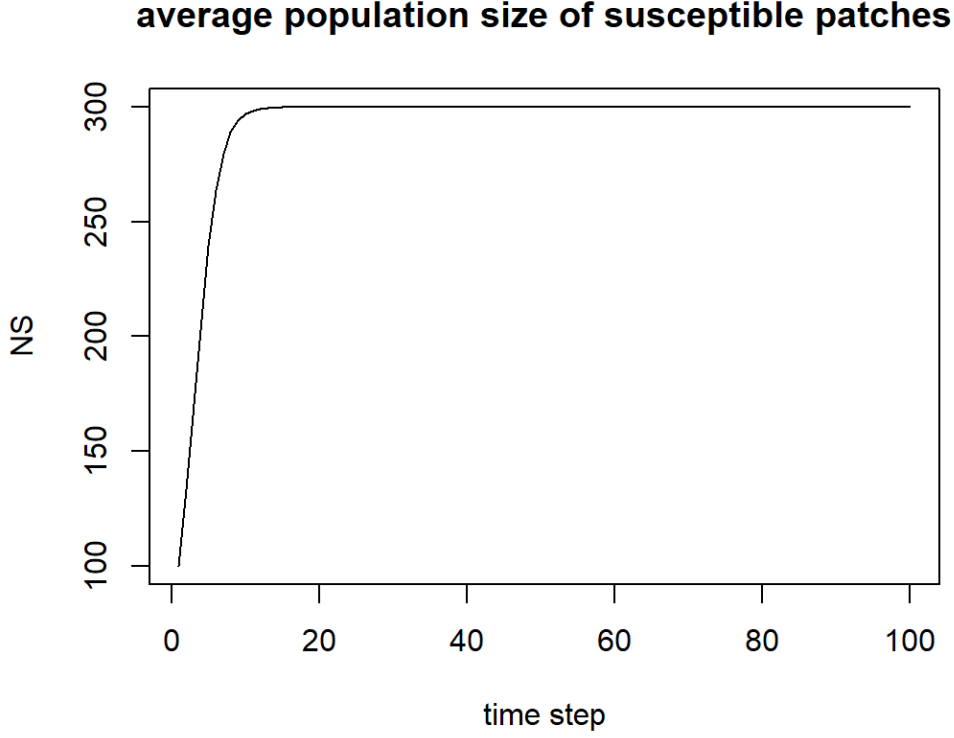


Figure 3: DFE

4 invasion analysis

$$J_{f_m} = \begin{pmatrix} 1 & 0 & 0 \\ -c(N_S - N_I) & 1 - c(1 - p) & c(1 - p) \\ 0 & cp & 1 - cp \end{pmatrix}$$

$$J_{f_i} = \begin{pmatrix} 1 + B - 2Bp & p(1 - p) \frac{\partial B}{\partial N_I} & p(1 - p) \frac{\partial B}{\partial N_S} \\ -\frac{B(N_S - N_I)}{(1 + B(1 - p))^2} & \frac{1 + B(1 - p) - (N_I - N_S)(1 - p) \frac{\partial B}{\partial N_I}}{(1 + B(1 - p))^2} & 1 - \frac{1 + B(1 - p) - (N_I - N_S)(1 - p) \frac{\partial B}{\partial N_S}}{(1 + B(1 - p))^2} \\ 0 & 0 & 1 \end{pmatrix}$$

$$J_{f_e} = \begin{pmatrix} P_1 & p \frac{\partial P_1}{\partial N_I} & 0 \\ 0 & 1 - zD & 0 \\ J_{31} & J_{32} & \frac{1 - p}{1 - P_1 p} \end{pmatrix}$$

where

$$J_{31} = \frac{(-N_S + N_I(1 - P_1)(1 - zD))(1 - P_1 p) + ((1 - p)N_S + (1 - zD)(1 - P_1)pN_I)P_1}{(1 - P_1 p)^2}$$

$$J_{32} = \frac{p(1 - zD)(-\frac{\partial P_1}{\partial N_I}N_I + 1 - P_1)(1 - P_1 p) + ((1 - p)N_S + (1 - zD)(1 - P_1)pN_I)p \frac{\partial P_1}{\partial N_I}}{(1 - P_1 p)^2}$$

$$J_{f_g} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 + r - \frac{2rN_I}{K} & 0 \\ 0 & 0 & 1 + r - \frac{2rN_S}{K} \end{pmatrix}$$